

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Declared actuator modes bound mean coupling by installed energy, giving a necessary resource condition for the scalar synchronization threshold. Phase encoders shared on overlaps convert coherence into a vector-content disagreement bound. Compatible spatial measures glue into one state; a contracting reconstruction map that fixes it makes it the unique fixed point. Reconstruction across a declared state family additionally requires distinguishable encodings, and local communication limits timely responses to interventions. Lean 4 checks these results and two compositions, each with eight connecting hypotheses and additional structures. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. The awakening hypothesis links extracellular geometry to increasing coupling. Finite ramps delay stationary onset, while coupling increase, diffusion decrease and a matched common drive can share stationary phase distributions. Testing requires calibrated physical identifications, resolved

dynamics and independent measurements of content agreement and self-encoding.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Consider reaching for a cup. A visual description of its location and a description of the hand's relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation.

21 This example motivates the requirements; the mathematical constructions
22 use spatial measures and fields in real normed vector spaces, whose identifi-
23 cation with perceptual content is a further modelling task.

24 The results specify both connections and separations between these re-
25 quirements. A declared actuator’s stored energy bounds its mean coupling,
26 giving a necessary energy condition for the scalar coherence threshold. Equal
27 phases permit incompatible local descriptions when content is independent
28 of phase; when a common Lipschitz encoder reads phase, coherence bounds
29 disagreement. Exact compatibility yields a unique global extension, and ap-
30 proximate compatibility bounds the variation among specified selections. A
31 contracting reconstruction map has a unique fixed point, with a quantita-
32 tive error bound, but identifying that point with the state formed from local
33 descriptions requires accuracy at that state.

34 Awakening gives the physical hypothesis a concrete test. We propose that
35 changing extracellular geometry increases coupling through the synchroniza-
36 tion threshold. The stationary onset is square-root, but finite ramp speeds
37 delay recovery. Moreover, increasing coupling, decreasing diffusion and a
38 matched common drive can produce identical stationary phase distributions.
39 A separate plasticity control preserves coherence while losing alignment with
40 its diagnostic environmental template. These results identify measurements
41 and mechanisms that a successful coordination model must supply.

42 The article follows the three requirements and their tests (Fig. 1). Ther-
43 modynamic constraints specify the costs of candidate implementations, in-

cluding agents that act on their future inputs. The detailed feedback, learning and resource models have distinct state spaces and assumptions; their consequences do not together instantiate a cortical agent. Section 6 summarizes the conditional composition, and Supplemental Material provides the constructions, witnesses and numerical controls.

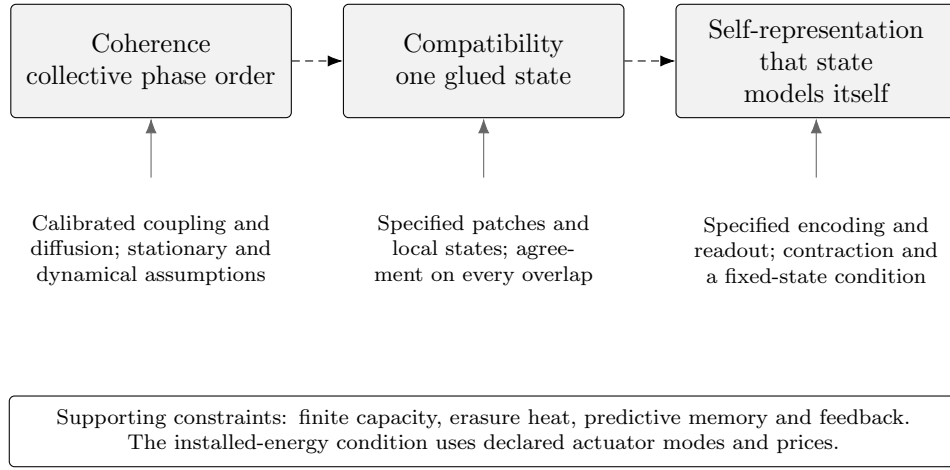


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

49 2 Coherence: physical coupling and its re- 50 source condition

51 2.1 Dynamics and the stationary threshold

52 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

53 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
54 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
55 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
56 parameters.

57 For identical frequencies in their rotating frame, the positive normalized
58 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

59 where I_n are modified Bessel functions and ψ is the collective phase. The for-
60 mal derivation uses the absence of stationary current for a periodic gradient
61 drift to obtain this density, then verifies normalization and self-consistency.
62 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one
63 positive solution coexists with zero and increases strictly with K . Its onset

64 is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

65 Figure 2 shows this stationary branch.

66 Linearization about the uniform density gives rate $K/2 - D$ for the first
67 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
68 exactly above threshold. About the coherent density, the quadratic form
69 is strictly dissipative transverse to collective rotation, whose rate is zero.
70 These linear estimates concern the limiting equation. Relating them to finite
71 stochastic trajectories requires a dynamical mean-field limit and branch selec-
72 tion; exchange symmetry alone does not establish the needed independence
73 (Sznitman, 1991).

74 **2.2 Installed energy bounds mean coupling**

75 A spatial model replaces the finite sum by integration against a kernel. One
76 declared implementation carries response modes with continuous profiles ϕ_i ,
77 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

78 The same configuration sets the reciprocal kernel and stored energy. An
79 executed transition changes the kernel through its changed occupancies and
80 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
81 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-

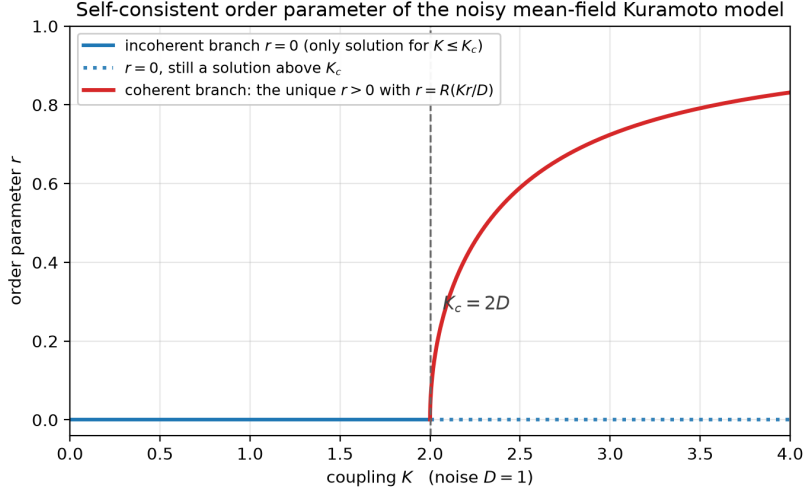


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

82 pair reconstruction on refining spatial partitions converges to its product-
 83 measure integral under the specified compactness and sampling assumptions;
 84 the refinement index is independent of the number of control updates.

85 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 86 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 87 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

88 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 89 coherent stationary branch and any growing Fourier mode about the uniform
 90 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 91 is necessary: energy above it need not produce sufficient coupling because
 92 the bound can be loose.

93 The quantity constrained is installed energy. Dissipated heat does not
 94 determine it, and κ is supplied by the mode profiles, prices and substrate
 95 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 96 tial dynamics requires a justified reduction. Identifying the installed kernel
 97 with a cortical field is a separate physical commitment.

98 2.3 Physical identification and calibration

99 The cortical hypothesis identifies part of effective coupling with endogenous
 100 extracellular electromagnetic interactions. Fields can modulate spike timing
 101 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
 102 propose coordination of distributed neural information (Pinotsis & Miller,
 103 2022; McFadden, 2020). Phase organization under propofol motivates study-
 104 ing this regime (Bastos et al., 2021), but does not measure coupling.

105 The formal field predicate requires a jointly continuous kernel, a normal-
 106 ized substrate, coupling equal to the kernel average, and no single site pro-
 107 viding more than half that strength. Toy constructions satisfy it. A cortical
 108 realization must specify the scale at which the kernel and its reduction ap-
 109 ply: calcium imaging reports different state-dependent modularity patterns

110 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

111 Finitely many components do not by themselves exclude continuum cou-
112 pling. Spreading a finite coupling matrix over cells of positive measure pre-
113 serves both total coupling weight and phase correlation exactly. Zero cou-
114 pling for finite point support under an atomless measure concerns that repre-
115 sentation of the support. Separately, a wiring constraint forces a correlation
116 disadvantage when it excludes a better-correlated pair. These comparisons
117 supply neither energy prices nor a biological–digital distinction; the supple-
118 ment gives the constructions and controls.

119 A spike-time shift alone supplies no coupling rate. The dimensionless
120 product of population size N , timing sensitivity s , field amplitude E and
121 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \tag{6}$$

122 This rate describes how induced timing shifts accumulate into drift. Nei-
123 ther it nor the installation factor κ follows from a heat bound. Testing the
124 physical threshold requires independent estimates of the interaction kernel,
125 its aggregation and diffusion in the same cortical regime.

126 **3 Testing coherence through recovery after** 127 **awakening**

128 The proposed control variable is extracellular geometry. Sleep-associated
129 changes in extracellular space (Xie et al., 2013) motivate coupling that evolves
130 more slowly than phase dynamics. Extracellular volume fraction is approxi-
131 mately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the sleep
132 study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz,
133 cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour,
134 2017). At fixed transmembrane current density, an effective-medium de-
135 scription predicts that a larger extracellular fraction raises conductivity and
136 reduces field amplitude.

137 The coupling-gated hypothesis consequently predicts that extracellular
138 contraction on awakening raises E and, holding the other factors in Eq. (6)
139 fixed, raises effective coupling. This argument supplies a direction; the
140 geometry–coupling relation, its magnitude and its role in human recovery
141 require measurement.

142 If $K(t)$ crosses K_c while the distribution tracks the coherent stationary
143 branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (7)$$

144 The crossing predicts latency only when coupling is measured independently

145 of coherence. Critical slowing down restricts branch tracking near threshold,
 146 and finite populations fluctuate below it. An observable onset therefore needs
 147 a criterion above that fluctuation floor.

148 **3.1 What a recovery distribution can identify**

149 Equation (2) gives a concentration–coherence relation with no fitted shape
 150 parameter. Estimate r from circular resultant length and a from density
 151 shape: inverting R from the measured r would make agreement automatic.
 152 Even independently estimated agreement identifies a statistical family shared
 153 by different mechanisms.

154 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 155 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 156 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 157 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 158 and diffusion D_0 have that same density. Their stationary distributions co-
 159 incide window by window despite different causes.

160 Resolved dynamics supply additional information. In the ideal model with
 161 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 162 increment means constrain drift. Separating endogenous coupling from a
 163 matched common drive additionally requires a measured drive or controlled
 164 perturbation; identifying electromagnetic coupling requires evidence specific
 165 to that interaction. Supplemental Material gives the equivalence construction
 166 and its numerical check.

167 **3.2 Finite ramps change the observable onset**

168 The ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling
169 speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after
170 threshold. Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape
171 within the fastest window. The three slower ramps have mean coupling
172 delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents run from 0.898 to
173 0.417, compared with 0.443 to 0.448 for the exact stationary branch on the
174 same grids. At the fastest speed the ensemble mean never reaches the fit
175 window; 0.964 is a local log-slope of a delayed trajectory.

176 This delayed passage through a noisy bifurcation (Berglund & Gentz,
177 2002) makes the apparent exponent depend on ramp speed and fitting win-
178 dow. A deviation from the square-root law tests the adiabatic approximation
179 where branch tracking is independently justified. The sweep does not bracket
180 the specified deviation-based critical ramp speed.

181 Additional controls address integration bias, population size and spatial
182 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
183 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
184 rameters and controls in Supplemental Material; they supply no cortical cal-
185 ibration.

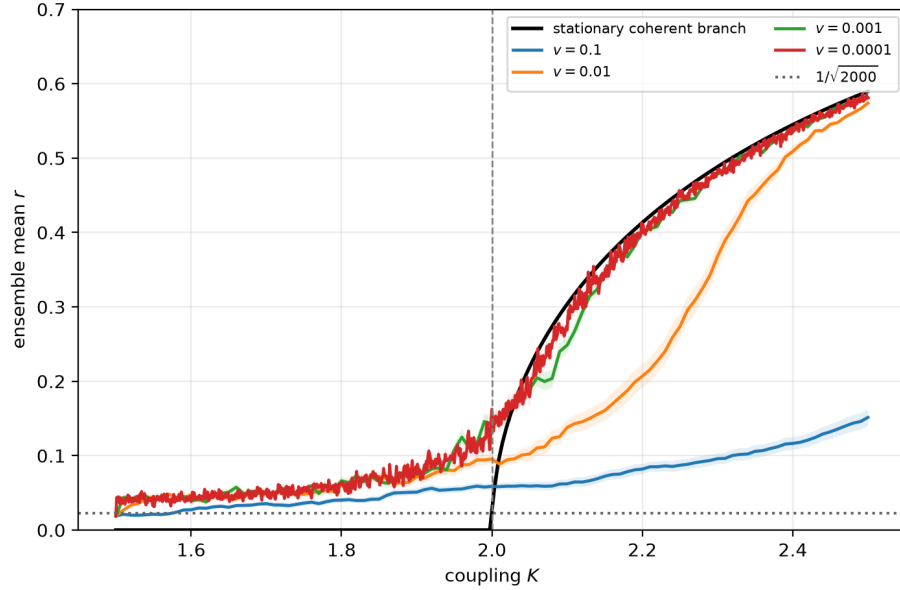


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

186 3.3 What the estimators produce on EEG

187 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
 188 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
 189 tages produce different residuals. The bipolar results are compatible with the
 190 Bessel relation, but their concentration range cannot distinguish it from its
 191 linear approximation at the observed scatter. Temporal pooling changes the
 192 estimand, and sedation recordings leave the awakening hypothesis untested.

193 An informative dataset needs broader concentration, calibrated estima-

194 tors and an observation model for dependence, source mixing and reference
 195 choice. Public mouse calcium imaging provides thousands of spatially lo-
 196 cated neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but
 197 its 7.65 Hz frame rate is below the phase-estimation band used here. Site
 198 count, state coverage and bandwidth are separate requirements.

199 **3.4 A protocol for the awakening experiment**

200 The physical test has six requirements:

- 201 **1. Coupling and diffusion in the same units.** Estimate them from re-
 202 solved phase dynamics, separating heterogeneity and observation noise.
 203 Measure the drive or perturb the system to distinguish endogenous cou-
 204 pling.
- 205 **2. Geometry calibrated to coupling.** Validate the geometry observ-
 206 able and its relation to effective coupling, with uncertainty. The con-
 207 ductivity argument supplies only the expected direction.
- 208 **3. Adequate phase measurements.** The proposed spatial estimator
 209 uses at least 100 sites, resultant length for r , and unweighted regres-
 210 sion of log density on centered cosine for a . Calibrate this numerical
 211 minimum under the actual dependence and concentration range.
- 212 **4. An independent recovery endpoint.** Measure behavioral recovery
 213 separately from phase organization.

214 **5. A predeclared onset criterion.** Set a coherence threshold above the
215 finite-size floor and account for ramp speed.

216 **6. Matched model comparison.** Compare coupling increase, diffusion
217 change, common drive and alternative recovery models under the same
218 observation and filtering assumptions.

219 Failure of the concentration–coherence relation over an informative range
220 constrains the density model. Failure of the calibrated geometry–coupling
221 association constrains the physical gate. Better prediction of independent
222 recovery by a competing mechanism counts against its proposed role. A
223 favorable outcome still leaves the content of coordinated activity to be ex-
224 plained.

225 **4 Compatibility: from coordinated activity** 226 **to a global description**

227 **4.1 Why common timing is insufficient**

228 For overlapping patches U and V , compatibility means that restricting either
229 local description to $U \cap V$ gives the same object. Descriptions can agree on
230 shared quantities while contributing different information elsewhere. For the
231 spatial measures used here, compatible restrictions give a unique measure on
232 the covered domain: the gluing property formalized by a sheaf.

233 In the cup example, visual and reaching descriptions must express a
 234 shared quantity, such as cup location, in common coordinates. Restriction
 235 retains that quantity while dropping details particular to each description.
 236 Agreement permits surface appearance and hand configuration to contribute
 237 to one situation. The mathematical counterexample assigns the same phase
 238 to every patch while two local mass profiles disagree on a shared site. Those
 239 profiles have no common extension. Because they are chosen independently
 240 of phase, synchrony places no constraint on their disagreement.

241 This distinction limits the biological interpretation. Receptive territories
 242 and content domains must be related to spatial patches, and the descriptions
 243 must support the required extension. Matching overlap marginals of de-
 244 coded probability laws need not determine a unique global joint distribution
 245 (Abramsky & Brandenburger, 2011). The spatial-measure result supplies a
 246 precise construction; applying it to perceptual content requires a justified
 247 content model.

248 **4.2 When coherence does constrain content**

249 A declared relation between phase and content gives a positive result. Let
 250 contents take values in a real normed vector space E . Patch i reads an
 251 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 252 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 253 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 254 sites; their equality on overlaps is a supplied relation between descriptions of

255 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 256 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 257 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (8)$$

258 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 259 agree exactly. The factor N makes this estimate potentially weak in large
 260 populations; it supplies a pointwise guarantee from a population average. A
 261 shared scalar encoder is a special case. Dropping the overlap relation permits
 262 disagreement even at perfect locking. A common nonexpansive linear projec-
 263 tion preserves agreement, but agreement of a projection does not establish
 264 agreement of the full vector.

265 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 266 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 267 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 268 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (9)$$

269 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 270 duce an initial mismatch. Approximate observations leave a nondecaying
 271 term in the error bound. Taking a uniform bound on Eq. (8) connects this
 272 update to the phase process. These are mechanisms for specified vector con-
 273 tents in shared coordinates; their neural meaning and relevance to experience

274 remain to be established.

275 **4.3 What approximate agreement determines**

276 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
277 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
278 Weights that vanish outside their patches and sum to one at each site select
279 a global profile by weighted averaging. The selection lies within ε of each
280 patch there, and any two such selections differ by at most ε , independently
281 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
282 $L\varepsilon$ across selections.

283 At zero tolerance, the selection is the unique exact extension. At every
284 positive tolerance there are families with no exact extension; selection then
285 gives an approximation whose dependence on weights is bounded. Supple-
286 mental Material gives a three-site witness attaining the bound. The mass-
287 profile construction uses the scalar case of the content dynamics and addi-
288 tionally requires nonnegativity of the initial profiles and observations. The
289 general vector result supplies no identification with the spatial-measure sheaf.
290 Neither construction selects the neural cover or decoder.

291 **4.4 Agreement after a change takes communication**

292 Snapshot compatibility does not specify how quickly descriptions respond
293 when a shared quantity changes. In a synchronous network whose updates

294 read only the site's own state and declared neighbours' messages, two initial
 295 configurations agreeing on a site's causal past through T rounds produce the
 296 same state there at round T . If two fresh interventions outside that past
 297 require reports separated by more than 2ε , no report from that site can be
 298 within ε of both. This follows from the reconstruction bound in Section 5.2.

299 A three-site delay line witnesses failure after one round and exact report-
 300 ing after two; cutting the path prevents delivery at every round. A report
 301 may still happen to be correct about one fixed world. The bound concerns
 302 responsiveness across interventions. Applying it requires a justified commu-
 303 nication graph, observation paths and deadline.

304 **4.5 Coherent adaptation can lose environmental align-** 305 **ment**

306 An adaptive coupling rule might supply representational structure, but its
 307 objective must reward the relevant agreement. For symmetric coupling, pair-
 308 wise sine terms cancel in the population mean drift. Minimizing squared drift
 309 at that fixed mean therefore rewards equalizing drift across nodes. In the
 310 frequency-cluster control, this favors coupling between unlike-frequency clus-
 311 ters, whereas the diagnostic environmental template rewards coupling within
 312 clusters.

313 For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (10)$$

314 The formalization constructs the drift operator A and adjoint, obtains $\nabla\sigma =$
 315 $(2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This
 316 functional differs from thermodynamic entropy production: at a coherent sta-
 317 tionary density the probability current vanishes while phase-averaged squared
 318 drift is positive.

319 The finite control co-evolves noisy phases and symmetric, nonnegative
 320 couplings at fixed total resource. Its template is a diagnostic covariance of
 321 latent signals, never a training input. Across three seeds, second-half order
 322 remains between 0.9167 and 0.9222, while the within-cluster to between-
 323 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
 324 Random-update, relabelled-gradient and frozen-kernel controls are reported
 325 in Supplemental Material.

326 The immediate post-update objective reduction disappears when aver-
 327 aged over complete update intervals. A prespecified grid of 6 rate–cadence
 328 combinations contains no case meeting the 5% sustained-reduction criterion
 329 while retaining coherence and losing alignment. The finding is coherent adap-
 330 tation away from the specified structure, with no demonstrated sustained cost
 331 reduction.

332 Kernel alignment and overlap compatibility are different observables.

333 This control addresses acquisition of its designated template under its par-
 334 ticular objective. Learning models can explicitly include data in an objective
 335 (Corberi et al., 2026); an account of agreement about the cup must specify
 336 the content its learning rule is meant to preserve.

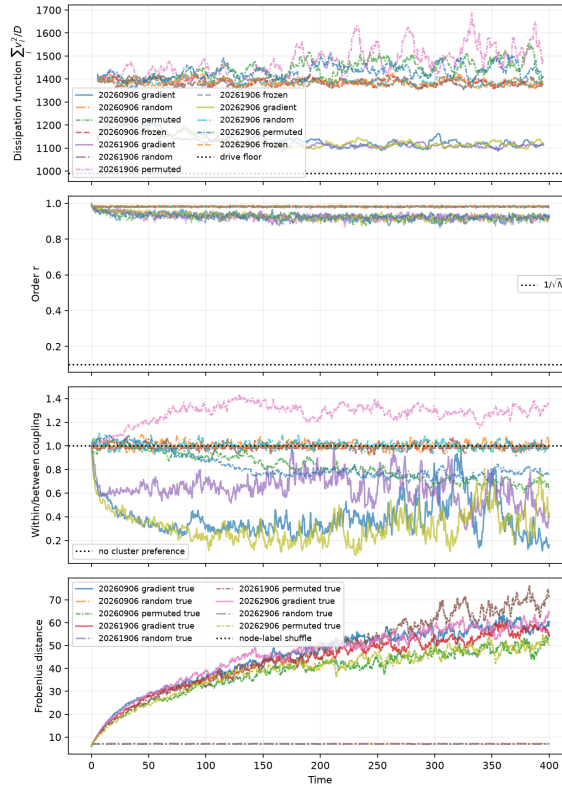


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

337 4.6 What must be specified and measured

338 A neural test needs independently identified domains, local descriptions and
339 restriction maps. Two decoders should estimate a shared content on over-
340 lapping domains before their agreement is examined. Discrepancy requires
341 a null under decoding noise and controls using deliberately compatible and
342 incompatible examples. Shared priors can manufacture agreement, while
343 regional bias can manufacture disagreement. A phase-agreement estimator
344 would measure the variable whose explanatory limits are at issue.

345 Choosing the cover also chooses a candidate subject: trivial descriptions
346 can agree while carrying little content, and a partition can conceal disagree-
347 ment. The useful choice must therefore be constrained by what neural ac-
348 tivity encodes. The exact composition assumes compatibility of its supplied
349 cover. The content mechanisms above identify additional hypotheses under
350 which agreement can arise; they leave the empirical choice of content and
351 the connection to that cover explicit.

352 5 Self-representation: a state that models it- 353 self

354 5.1 The reflexive construction

355 Compatibility gives a global state. Reflexivity relates it to an encoding within
356 that state. Equip global sections with a complete metric. If the additional

357 restriction-resonance condition identifies encoding with restriction to a region
 358 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (11)$$

359 The region and readout are fixed across the state space. If F contracts and
 360 fixes the section s_* formed from the local descriptions, Banach's theorem
 361 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 362 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 363 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 364 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 365 obligation.

366 In the cup example, the global state includes the represented scene and the
 367 perceiver's relation to it. A local encoding must retain enough information
 368 for the readout to recover that modelled state; cup position alone would be
 369 insufficient if hand configuration and viewpoint could not be reconstructed.
 370 Here *prediction* names a reconstruction map. Interpreting it as temporal
 371 neural prediction requires dynamics.

372 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 373 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (12)$$

374 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 375 fixed positive ε and τ . Exact uniqueness survives while the error guarantee

376 weakens. A biological interpretation must specify the represented variables,
 377 metric and acceptable fidelity; changing readouts need a further model.

378 5.2 Reconstruction across a family of states

379 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 380 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 381 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (13)$$

382 Thus M relevant states pairwise more than 2ε apart require at least M
 383 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 384 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 385 existing self-prediction map.

386 This is an additional accuracy criterion on a family, with no requirement
 387 for multiple exact fixed points. A blind encoder or constant readout fails it
 388 for separated states despite having a unique fixed point. The family, metric
 389 and tolerance are inputs; interpreting the code count as physical memory
 390 capacity requires a hardware model.

391 5.3 Why connect these properties to unity and a min- 392 imal self?

393 We propose the state glued from compatible local descriptions as a correlate
394 of conscious unity, and its accurate internal reconstruction as a correlate of
395 a minimal self. The motivations are that different contents belong to one ex-
396 perience situation, and that the situation is experienced from a perspective.

397 Compatibility expresses agreement about shared quantities while allow-
398 ing different contributions. It also permits a description of uncertainty or
399 conflicting judgments: apparent cup position and judged cup position can
400 be represented as distinct quantities within a consistent account of their dis-
401 crepancy. Defining which quantities are shared is substantive. A choice made
402 solely to secure agreement would deprive the proposal of empirical force.

403 The minimal-self proposal concerns the perceiver's position and condition
404 in that situation: where the cup appears relative to a body that can reach it.
405 Such an encoding could be implicit and need carry no autobiographical nar-
406 rative. Its proposed necessity is philosophical: even an unreflective episode
407 has a perspective. The gluing theorem alone requires no self-encoding.

408 Reconstruction also needs a functional interpretation. A reconstructable
409 recording need never be used to distinguish changes in a system from changes
410 in its surroundings. Altering the encoded hand or viewpoint while holding
411 the external scene fixed should change estimates of the perceiver's relation
412 to the cup as the readout predicts. Whether that represented perspective is

413 experienced is a further question.

414 **5.4 Selecting and testing the construction**

415 Both uniqueness results concern specified objects. They choose neither the
416 cover nor the encoding, and do not select between candidate subjects. Inte-
417 grated information theory treats exclusion explicitly (Tononi et al., 2016); the
418 present construction leaves selection to independent evidence about neural
419 organization.

420 A constant readout returning a predetermined state contracts and has a
421 fixed point while discarding its encoding. It fails the functional sensitivity
422 required above. Local descriptions must predict the contents assigned to
423 their domains; encoding and readout must be specified before testing their
424 responses over a declared range.

425 A test of the full proposal must separately measure coupling relative to
426 diffusion, content compatibility and self-reconstruction, alongside an inde-
427 pendent endpoint for experience. Meeting all three without a unified episode
428 would challenge sufficiency. A unified episode with incompatible descrip-
429 tions would challenge the unity identification; one lacking the specified self-
430 encoding would challenge the reflexive requirement. Awakening, dreaming
431 and dissociative states offer possible settings, but absence of report cannot
432 by itself define absence of experience.

433 The compatibility observable is currently a specification, and the neural
434 self-encoding has yet to be identified. Gluing and fixed-point conditions

435 permit multiple substrates; the electromagnetic commitment enters through
 436 physical calibration. The proposal concerns which state is conscious and
 437 leaves unexplained why those relations would be accompanied by experience.

438 **6 Thermodynamics and composition**

439 The phase and representation results constrain a candidate implementation
 440 through different quantities: coupling, content disagreement and reconstruc-
 441 tion error. Thermodynamics additionally constrains its physical operations.
 442 Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on
 443 specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker,
 444 2000). A separate boundary result concerns a field reaching disconnected
 445 vacuum components (Kibble, 1976). Their physical connections are inputs
 446 to the composition below.

447 **6.1 Feedback changes the information constraint**

448 For passive memory X_t driven by an environmental signal S_t , the predictive
 449 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (14)$$

450 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 451 et al., 2012). Zero retained information also satisfies it, so a task and a
 452 learning rule require additional specification. Relating this bound to Lan-

453 dauer heat requires a common thermal scale and budget, and memory waste
 454 bounded by erased entropy.

455 An acting agent changes the signal it will receive. Specify policy, world,
 456 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{15}$$

457 The second conditional term vanishes in the passive model. With feedback,
 458 the signed difference can be negative as action creates correlations with the
 459 future. A causal reading requires an environmental state sufficient to describe
 460 the relevant dynamics.

461 For a controlled substep, the active thermodynamic bound instead uses
 462 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 463 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 464 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 465 2013). Every forward path of positive probability must have a positive-
 466 probability reverse for this finite identity. A missing reverse gives infinite
 467 divergence. The supplement develops the support conditions and composes
 468 actuation with memory updates on their shared intermediate law.

469 6.2 Learning and repeated operation need explicit re- 470 sources

471 The finite examples distinguish selecting a feasible policy, relaxing a register
472 toward an encoded reward, and learning an unknown task from observa-
473 tions. The observational learner acquires correlation with an environmental
474 parameter while its action marginal stays uniform. A continuing model exe-
475 cutes actuation, observation-driven learning and reset in sequence, preserving
476 above-chance performance at a positive work cost per cycle.

477 One observation channel is derived from a declared phase configuration:
478 sample an oscillator uniformly and report its phase through a specified finite
479 readout. Symbol probabilities are population fractions. An informative four-
480 oscillator sensor and an uninformative control have the same $r^2 = 1/4$ at
481 every parameter and action. Under the same update rule, their expected
482 rewards are respectively $3/4 - (1/4)2^{-n}$ and $1/2$ after n updates. Their
483 difference lies in the phase-world relation and readout, which coherence alone
484 does not specify.

485 Attaching the learner's register to the declared actuator of Section 2.2
486 ties reward, installed kernel and work to that same executed law. In this
487 exact witness, $U_n = 3/4 - (1/4)2^{-n}$ and mean installation work per update is
488 $(1+\log 3)2^{-n}/8$; cumulative work stays within $(1+\log 3)/4$. The control keeps
489 $U_n = 1/2$ and spends zero mean work. At the witness's $\kappa = 4$ and $D = 1$,
490 the control remains on the exclusion line while one learning update raises

491 the learner above it. Clearing that necessary energy condition establishes
 492 no evolving coherent field: phase dynamics, installation and sensing are not
 493 closed into a feedback trajectory.

494 For that model, common energy E , thermal scale θ , local detailed balance
 495 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (16)$$

496 where expected cumulative work stays within the supplied allowance B at
 497 every prefix. This bounds expectations. Separate constructions track a store
 498 on each trajectory, include finite sources, charge sensor clearing and prepend
 499 preparation. They specify when a mean allowance represents an available
 500 resource; an expected budget alone permits overdrawn paths.

501 These constructions establish costs for their declared processes. A finite
 502 source with a positive net cost permits only finitely many funded cycles,
 503 while continuing operation requires replenishment. Register–bath witnesses
 504 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply
 505 automatically.
 507

508 6.3 What the composition establishes

509 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 510 cept eight connecting implications (Table 1). In the active branch, a named
 511 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (17)$$

512 with B the register’s heat. The upper allocation follows from a ledger whose
 513 operations deliver that total heat to one reservoir and whose other operations
 514 do not increase joint entropy. Those ledger assumptions are physical inputs.
 515 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (11).

Joint witnesses combine a register, boundary, refining mesh and small

523 spatial cover; active witnesses add controlled thermal updates and a priced
524 installation. They establish satisfiability under their stated identifications.
525 Mesh convergence follows from refinement data, and the witnesses establish
526 no common cortical mechanism. Supplemental Material maps the claims to
527 their Lean declarations and records the counterexamples. The development
528 declares no additional axioms beyond its libraries' foundational axioms.

529 **7 Discussion**

530 The framework distinguishes the physical conditions for coherent activity
531 from the relations needed to interpret it as one internally represented sit-
532 uation. Its positive connections depend on specified mechanisms: installed
533 modes relate energy to mean coupling, phase encoders shared on overlaps
534 relate coherence to vector-content agreement, and an observation update
535 propagates that agreement into retained fields. Their assumptions identify
536 what a cortical model must establish.

537 The separation results delimit what would count as evidence. A station-
538 ary recovery distribution leaves coupling, diffusion and a matched common
539 drive indistinguishable. Synchronous patches can carry incompatible phase-
540 independent descriptions. Coherent adaptation under the tested squared-
541 drift objective can lose environmental alignment. Each result concerns the
542 stated model and observable; none determines cortical learning from ther-
543 modynamics or diagnoses experience from synchrony alone.

544 For the awakening hypothesis, the next requirement is a calibrated
545 geometry–coupling relation and resolved phase dynamics, with an indepen-
546 dent recovery endpoint. The installation bound adds a necessary resource
547 condition where its hardware representation applies. It does not calibrate
548 cortical mode prices or replace evidence that the physical field operates
549 above threshold.

550 For the representational connection, the informative experiment is
551 whether a controlled change in coordination alters independently decoded
552 content agreement. Decoders and restriction maps must first recover compat-
553 ible and incompatible cases under realistic noise. Perturbations then need
554 controls for sensory information and decoder quality. The phase-encoder
555 theorem supplies one mechanism to test, but shared overlap encoders, com-
556 mon coordinates and population-size dependence limit how directly it can
557 describe neural content. Equal coherence can also accompany different task
558 information, as the phase-sensor control demonstrates.

559 The self clause requires a candidate encoding whose perturbation changes
560 the represented scene, body or viewpoint as its readout predicts. Approx-
561 imate gluing and reconstruction make fidelity an explicit parameter. They
562 leave the cover, metric, encoder and required resolution to be justified inde-
563 pendently; uniqueness within these choices does not choose a subject.

564 Integrated-information, global-workspace and predictive-processing ac-
565 counts organize unity, access and modelling differently (Tononi et al., 2016;
566 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal con-

567 tributes a candidate physical interaction and conditional relations between
568 local content and self-encoding. Its explanatory value depends on identifying
569 those relations in neural activity and showing that they matter for experi-
570 ence. In the cup example, the challenge is to explain how coordination helps
571 maintain one represented situation and how the perceiver’s place within it
572 contributes to the episode.

573 **Data and code availability**

574 The Lean development, simulation scripts, tests and saved numerical sum-
575 maries are maintained in the manuscript source repository. Publication sim-
576 ulation values are regenerated from saved summaries without rerunning pro-
577 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
578 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
579 in Supplemental Material. The source repository contains the reproducible
580 analysis and does not redistribute the raw recordings.

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584 Use of AI tools

585 AI assistance to this work was substantial and is not separable by task.
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588 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
589 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
590 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
591 they contributed to the Lean development, the simulation and analysis code,
592 and the drafting and revision of both documents. Because the tools were
593 used in combination and in alternation, no section, proof, figure or result is
594 attributable to any one of them, and none is claimed to be. The author is
595 responsible for all content, including everything a tool produced. Machine
596 checking establishes the stated Lean results under their hypotheses; it does
597 not validate biological interpretations or replace scientific review of generated
598 code and prose.

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